

1 Solution — GIAM 3.1

Problem 2

1.1 Problem

Prove that 129 is odd.

1.2 Proof

Let $k = 64$. Then $k \in \mathbb{Z}$ and $2k + 1 = 2(64) + 1 = 129$. Hence 129 is odd. ■

That is the complete proof. The definition of *odd* is

$$n \text{ is odd} \iff \exists k \in \mathbb{Z}, n = 2k + 1,$$

so proving "129 is odd" requires exactly one thing: **exhibiting an integer k** that works. Producing $k = 64$ and checking the arithmetic discharges the obligation entirely.

1.3 Why One Line Suffices

The claim is existential, so the whole proof is: unpack the definition, then hand over the witness.

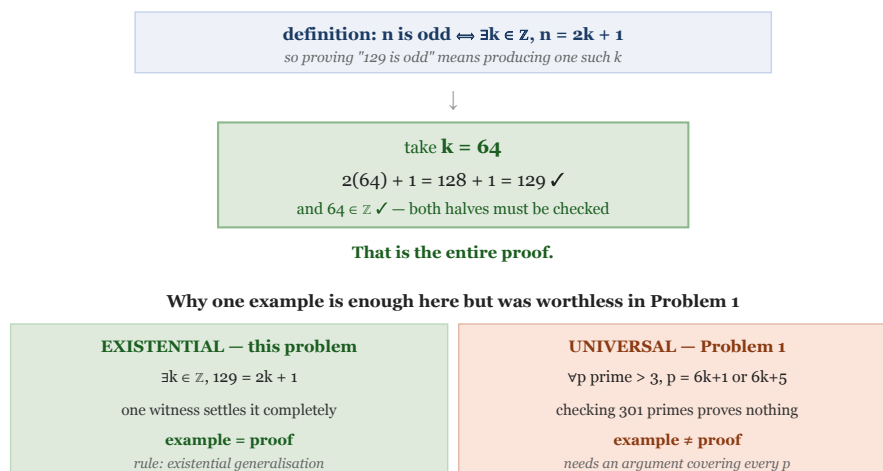


Figure 1.1: Top: the definition, n is odd if and only if there exists an integer k with n equal to $2k$ plus 1 — so proving 129 is odd means producing one such

k. Below, the witness: take k equal to 64; then 2 times 64 plus 1 equals 128 plus 1 equals 129, and 64 is an integer — both halves must be checked. That is the entire proof. Below that, a two-panel contrast titled ‘why one example is enough here but was worthless in Problem 1’. Left panel, **EXISTENTIAL**, this problem: there exists an integer k with 129 equal to $2k$ plus 1; one witness settles it completely, so example equals proof, by the rule of existential generalisation. Right panel, **UNIVERSAL**, Problem 1: for all primes p greater than 3, p is $6k+1$ or $6k+5$; checking 301 primes proves nothing, so example does not equal proof, and an argument covering every p is needed.

The rule of inference is **existential generalisation** (Problem 2.7.5): from the particular fact $129 = 2(64) + 1$ we infer $\exists k \in \mathbb{Z}, 129 = 2k + 1$. Nothing more is required, because the statement being proved *is* an existential.

This is worth contrasting sharply with the previous problem. There the claim was **universal** — “every prime $p > 3$ has the form $6k + 1$ or $6k + 5$ ” — and verifying it for all 301 primes below 2000 would have proved *nothing*, because a universal claim ranges over infinitely many cases. Here the claim is **existential**, and a single instance is not merely evidence but a complete proof.

1.4 Remark — Both Halves of the Witness Must Be Checked

The proof has two obligations, and it is easy to notice only the first:

1. **The equation holds:** $2(64) + 1 = 129$. ✓
2. **The witness is in the right set:** $64 \in \mathbb{Z}$. ✓

Skipping the second is a genuine error, not pedantry. Every number n satisfies $n = 2k + 1$ for *some* real k — take $k = (n - 1)/2$ — so without the requirement $k \in \mathbb{Z}$ the definition would make every number odd, including 128 (with $k = 63.5$). The integrality condition is the entire content of the definition. This is precisely the point flagged in Problem 2.7.5, where “ $k + j + 1$ is an integer” turned out to be a load-bearing line rather than a fussy one.

1.5 Remark — Trivial Problems Test Whether You Know the Definition

A problem this small can feel like a joke, but it is doing real work: it checks whether you can **write down what a word means** and act on it. A student who cannot prove "129 is odd" in one line is usually one who has an *intuition* for oddness ("it doesn't divide evenly," "it ends in 9") but has not internalised the formal definition $\exists k \in \mathbb{Z}, n = 2k + 1$.

The intuitions are not wrong, but they are not usable in a proof. "Ends in 9" is a fact about decimal notation, not about the integer, and would require its own justification. The formal definition is the only version that plugs directly into later arguments — which is why the very first proofs in a course are always of this shape: *unpack the definition, satisfy it literally*.

1.6 Remark — Proving the Negative Would Take More Work

Suppose instead the task were "prove 129 is **not even**." That is a *universal* claim in disguise — $\neg \exists m \in \mathbb{Z}, 129 = 2m$, equivalently $\forall m \in \mathbb{Z}, 129 \neq 2m$ — so no single witness will do. One would have to argue, for instance:

*If $129 = 2m$ for an integer m , then $m = 64.5 \notin \mathbb{Z}$, a contradiction. Alternatively, the quotient-remainder theorem gives $129 = 2 \cdot 64 + 1$ with remainder 1, and that remainder is **unique**, so no representation with remainder 0 exists.*

Notice how much heavier this is: it needs either a contradiction or an appeal to a uniqueness theorem. The asymmetry — existentials proved by exhibition, universals needing an argument — is the single most useful structural fact about proofs, and it is why identifying the *shape* of a claim (as Problem 1 asked) comes before choosing a technique.

1.7 Remark — Odd and Even Are Not Opposites by Definition

A closing subtlety. The definitions

$$n \text{ odd} \iff \exists k \in \mathbb{Z}, n = 2k + 1, \quad n \text{ even} \iff \exists m \in \mathbb{Z}, n = 2m$$

are stated independently — nothing in them says a number must be exactly one of the two. That every integer is *either* odd or even (and never both) is a **theorem**, following from the quotient–remainder theorem applied with divisor **2**: the remainder is **0** or **1**, and it is unique.

So the one-line proof above establishes that **129** is odd, but it does not by itself establish that **129** is not even — that needs the extra theorem. The distinction matters because in other settings the analogous dichotomy can fail; among the rationals, for instance, “odd” and “even” are simply not defined. Knowing which convenient facts are definitions and which are earned theorems is exactly the habit Chapter 3 is beginning to build.